

AAE 50700: Principles of Dynamics
Problem Set 5 Report

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(5.1) Orbital Dynamics with J_2 and Atmospheric Drag

(a): Given

Consider a spacecraft of mass m orbiting Earth perturbed by J_2 and atmosphere drag. Let the inertial frame be $\mathcal{N} : \{\hat{e}_x, \hat{e}_y, \hat{e}_z\}$ and write its Cartesian position as $\mathbf{r} = x\hat{e}_x + y\hat{e}_y + z\hat{e}_z, r = \|\mathbf{r}\|_2$. Expressing the potential energy for Earth gravity under J_2 in the Cartesian coordinates, we have

$$V(x, y, z) = -\frac{\mu m}{r} + \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2), \quad (1)$$

where μ is Earth's gravitational parameter and $r_o = R_{\text{earth}}$ is its equatorial radius.

We use the same model as considered in PS1 to model the drag, which assumes an inertially stationary, spherically symmetric atmosphere, given by

$$\mathbf{F}_D = m\mathbf{a}_D, \quad \mathbf{a}_D = -\frac{1}{2}C_D \frac{A}{m} \rho(r) v^2 \hat{\mathbf{v}}, \quad v := \|\mathbf{v}\|_2, \quad \hat{\mathbf{v}} := \frac{\mathbf{v}}{v}, \quad (2)$$

where $\mathbf{v} = \dot{\mathbf{r}}$, C_D is the drag coefficient, A is the reference area, and

$$\rho(r) \approx \rho_0 \exp \left[-\frac{r - 6978 \text{ km}}{H} \right]. \quad (3)$$

(a.1): Find

From the Lagrangian using the Cartesian coordinates as generalized coordinates, $\mathbf{q} = [x, y, z]^T$.

Analysis/Conclusions:

First recall that the kinetic energy T of an object in motion can be expressed by half the mass times the square of the velocity:

$$T(\mathbf{v}) = \frac{m}{2} \mathbf{v}^T \mathbf{v} = \frac{mv^2}{2} \quad (4)$$

Then, recall the Lagrangian L is the kinetic energy minus the potential energy $L = T - V$ (V given by the problem definition):

$$L(\mathbf{r}, \mathbf{v}) = \frac{mv^2}{2} + \frac{\mu m}{r} - \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \quad (5)$$

(a.2): Find

Derive the generalized force for Eq. (2) associated with $\mathbf{q} = [x, y, z]^T$.

Analysis/Conclusions:

It is known that virtual work δw is translational force $\mathbf{F}$ dotted with the Jacobian of position with respect to the generalized coordinates multiplied by the displacement of generalized coordinates. This also equals generalized force $\mathbf{Q}$ dotted with the same displacement:

$$\delta w = \mathbf{F}^T \frac{\partial \mathbf{r}}{\partial \mathbf{q}} \delta \mathbf{q} = \mathbf{Q}^T \delta \mathbf{q}. \quad (6)$$

Since in this case the generalized coordinates are the Cartesian position, $\mathbf{q} = \mathbf{r}$, the Jacobian is the identity matrix, and the generalized force is simply the applied force:

$$\delta w = \mathbf{F}^T \frac{\partial \mathbf{r}}{\partial \mathbf{r}} \delta \mathbf{r} = \mathbf{F}^T I \delta \mathbf{r} = \mathbf{F}^T \delta \mathbf{r} \Rightarrow \mathbf{Q} = \mathbf{F}. \quad (7)$$

So, from Eq. (2) there is only one force, which is the force of drag $\mathbf{F}_D$. This means the generalized force from that equation associated with $\mathbf{q}$ is $\mathbf{Q} = \mathbf{F}_D$. The mass terms cancel and simplify to get generalized force:

$$\mathbf{Q} = \mathbf{F}_D = m \mathbf{a}_D = m \left(-\frac{1}{2} C_D \frac{A}{m} \rho(r) v^2 \hat{\mathbf{v}} \right) = -\frac{1}{2} C_D A \rho(r) v^2 \hat{\mathbf{v}} \quad (8)$$

(a.3): Find

Derive the Lagrange's equations and write the result as the Cartesian vector EOM

$$\ddot{\mathbf{r}} = \mathbf{a}_{2B} + \mathbf{a}_{J2} + \mathbf{a}_D \quad (9)$$

Analysis/Conclusions:

We see here that this system has three degrees of freedom with no redundancy in coordinates, so $\mathbf{q}$ is a minimal coordinate set, and the system has no constraints on movement. Since there is also a non-conservative force (drag), the general form of Lagrange's equations with non-conservative general force may be used:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial L}{\partial \mathbf{q}} = \mathbf{Q}_{nc} \quad (10)$$

And since, $\mathbf{q} = \mathbf{r}$ and $\mathbf{F}_D$ is the only non-conservative general force:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{v}}} \right) - \frac{\partial L}{\partial \mathbf{r}} = \mathbf{F}_D \quad (11)$$

Then we compute the derivatives in Eq. (11) individually.

So first to solve $\frac{\partial L}{\partial \mathbf{v}}$:

$$\begin{aligned}\frac{\partial L}{\partial \mathbf{v}} &= \frac{\partial}{\partial \mathbf{v}} \left(\frac{mv^2}{2} + \frac{\mu m}{r} - \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= \frac{m}{2} \frac{\partial v^2}{\partial \mathbf{v}} + \frac{\partial}{\partial \mathbf{v}} \left(\frac{\mu m}{r} \right) - \frac{\partial}{\partial \mathbf{v}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= \frac{m}{2} 2\mathbf{v}^T + 0 - 0 = m\mathbf{v}^T\end{aligned}\tag{12}$$

Then to solve $\frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right)$:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \mathbf{v}} \right) = m \frac{d\mathbf{v}^T}{dt} = m\mathbf{a}^T\tag{13}$$

Then solve $-\frac{\partial L}{\partial \mathbf{r}}$:

$$\begin{aligned}-\frac{\partial L}{\partial \mathbf{r}} &= -\frac{\partial}{\partial \mathbf{r}} \left(\frac{mv^2}{2} + \frac{\mu m}{r} - \frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= -\frac{\partial}{\partial \mathbf{r}} \left(\frac{mv^2}{2} \right) - \mu m \frac{\partial r^{-1}}{\partial \mathbf{r}} + \frac{\partial}{\partial \mathbf{r}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\ &= 0 - \mu m \frac{\partial r^{-1}}{\partial \mathbf{r}} + \frac{\partial}{\partial \mathbf{r}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right)\end{aligned}\tag{14}$$

Then for readability, we do the remaining component derivatives individually:

$$\frac{\partial r^{-1}}{\partial \mathbf{r}} = \frac{\partial r^{-1}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} = -r^{-2} \hat{\mathbf{r}}^T\tag{15}$$

And the other component:

$$\begin{aligned}
&= \frac{\partial}{\partial \mathbf{r}} \left(\frac{\mu m J_2 r_o^2}{2r^5} (3z^2 - r^2) \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(\frac{\partial}{\partial \mathbf{r}} (3r^{-5} z^2) - \frac{\partial r^{-3}}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left(\frac{\partial r^{-5}}{\partial \mathbf{r}} z^2 + r^{-5} \frac{\partial z^2}{\partial \mathbf{r}} \right) - \frac{\partial r^{-3}}{\partial \mathbf{r}} \frac{\partial r}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left(\frac{\partial r^{-5}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} z^2 + r^{-5} \frac{\partial z^2}{\partial z} \frac{\partial z}{\partial \mathbf{r}} \right) - \frac{\partial r^{-3}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left(\frac{\partial r^{-5}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} z^2 + r^{-5} \frac{\partial z^2}{\partial z} \hat{\mathbf{e}}_z^T \frac{\partial \mathbf{r}}{\partial \mathbf{r}} \right) - \frac{\partial r^{-3}}{\partial r} \frac{\partial r}{\partial \mathbf{r}} \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left((-5r^{-6})(\hat{\mathbf{r}}^T) z^2 + 2r^{-5} z \hat{\mathbf{e}}_z^T I \right) - (-3r^{-4} \hat{\mathbf{r}}^T) \right) \\
&= \frac{1}{2} \mu m J_2 r_o^2 \left(3 \left((-5r^{-6})(r^{-1} \mathbf{r}^T) z^2 + 2r^{-5} z \hat{\mathbf{e}}_z^T I \right) - (-3r^{-4})(r^{-1} \mathbf{r}^T) \right) \\
&= \frac{3}{2} \mu m J_2 r_o^2 \left(-5r^{-7} z^2 \mathbf{r}^T + 2r^{-5} z \hat{\mathbf{e}}_z^T + r^{-5} \mathbf{r}^T \right) \\
&= \frac{3}{2} \mu m J_2 r_o^2 r^{-5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right) \\
&= \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right)
\end{aligned} \tag{16}$$

So then that gives:

$$\begin{aligned}
-\frac{\partial L}{\partial \mathbf{r}} &= -\mu m \left(-r^{-2} \hat{\mathbf{r}}^T \right) + \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right) \\
&= \frac{\mu m}{r^3} \mathbf{r}^T + \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right)
\end{aligned} \tag{17}$$

Which then plugging the above terms back into Eq. (11) gives:

$$m \mathbf{a}^T + \frac{\mu m}{r^3} \mathbf{r}^T + \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r}^T + 2z \hat{\mathbf{e}}_z^T \right) = -\frac{1}{2} C_D A \rho(r) v^2 \hat{\mathbf{v}}^T \tag{18}$$

Which then solved for acceleration $\mathbf{a} = \ddot{\mathbf{r}}$:

$$\begin{aligned}
m \mathbf{a} &= -\frac{\mu m}{r^3} \mathbf{r} - \frac{3\mu m J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z \hat{\mathbf{e}}_z \right) - \frac{1}{2} C_D A \rho(r) v^2 \hat{\mathbf{v}} \\
\mathbf{a} &= -\frac{\mu m}{r^3 m} \mathbf{r} - \frac{3\mu m J_2 r_o^2}{2r^5 m} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z \hat{\mathbf{e}}_z \right) - \frac{1}{2m} C_D A \rho(r) v^2 \hat{\mathbf{v}} \\
\mathbf{a} &= -\frac{\mu}{r^3} \mathbf{r} - \frac{3\mu J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z \hat{\mathbf{e}}_z \right) - \frac{1}{2m} C_D A \rho(r) v^2 \hat{\mathbf{v}}
\end{aligned} \tag{19}$$

Which means the Cartesian vector EOM are:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} - \frac{3\mu J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z\hat{\mathbf{e}}_z \right) - \frac{1}{2m}C_D A \rho(r)v^2\hat{\mathbf{v}} \quad (20)$$

Where:

$$\begin{aligned} \mathbf{a}_{2B} &= -\frac{\mu}{r^3}\mathbf{r} \\ \mathbf{a}_{J2} &= -\frac{3\mu J_2 r_o^2}{2r^5} \left(\left(1 - \frac{5z^2}{r^2} \right) \mathbf{r} + 2z\hat{\mathbf{e}}_z \right) \\ \mathbf{a}_D &= -\frac{1}{2m}C_D A \rho(r)v^2\hat{\mathbf{v}} \end{aligned} \quad (21)$$

Since these terms come from the two-body problem, J_2 perturbation, and drag acceleration components of the system respectively.

(b): Given

Now consider spherical coordinates $\mathbf{q} = [r, \phi, \theta]^T$, where r is radius, ϕ is azimuth about $\hat{\mathbf{e}}_z$, and θ is elevation measured from $\hat{\mathbf{e}}_x - \hat{\mathbf{e}}_y$ plane. The potential energy for gravity with J_2 is given by:

$$V = -\frac{m\mu}{r} \left[1 - J_2 \left(\frac{r_o}{r} \right)^2 P_2(\sin \theta) \right] \quad (22)$$

(b.1): Find

Form the Lagrangian using spherical coordinates.

Analysis/Conclusions:

The Lagrangian is once again found by kinetic minus potential energy $L = T - V$. The gravity potential including J_2 zonal harmonics was already given by Eq. (22), then the kinetic energy is once again given by:

$$T = \frac{mv^2}{2} \quad (23)$$

To express T in spherical coordinates, we write the position as $\mathbf{r} = r\hat{\mathbf{e}}_r$ in the spherical basis $\mathcal{S} : \{\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\phi, \hat{\mathbf{e}}_\theta\}$. Since this basis rotates relative to the inertial basis $\mathcal{N} : \{\hat{\mathbf{e}}_x, \hat{\mathbf{e}}_y, \hat{\mathbf{e}}_z\}$, we find $\mathbf{v}$ with the transport theorem, which requires the rotation of the spherical basis with respect to the inertial Cartesian basis, $\boldsymbol{\omega}_{\mathcal{S}/\mathcal{N}}$. We can represent this by considering $\dot{\phi}$ scaled by the $\hat{\mathbf{e}}_r$ and $\hat{\mathbf{e}}_\theta$ components of $\hat{\mathbf{e}}_z = \sin \theta \hat{\mathbf{e}}_r + \cos \theta \hat{\mathbf{e}}_\theta$, as well as $\dot{\theta}$ scaled by the negative direction of $\hat{\mathbf{e}}_\phi$:

$$\boldsymbol{\omega}_{\mathcal{S}/\mathcal{N}} = \dot{\phi} \sin \theta \hat{\mathbf{e}}_r + \dot{\phi} \cos \theta \hat{\mathbf{e}}_\theta - \dot{\theta} \hat{\mathbf{e}}_\phi \quad (24)$$

Which we then find $\mathbf{v}$ via the transport theorem as:

$$\begin{aligned} \mathbf{v} &= \frac{{}^S d}{dt} \mathbf{r} + \boldsymbol{\omega}_{\mathcal{S}/\mathcal{N}} \times \mathbf{r} \\ &= \frac{{}^S d}{dt} (r\hat{\mathbf{e}}_r) + \boldsymbol{\omega}_{\mathcal{S}/\mathcal{N}} \times (r\hat{\mathbf{e}}_r) \\ &= \frac{{}^S dr}{dt} \hat{\mathbf{e}}_r + r \frac{{}^S d\hat{\mathbf{e}}_r}{dt} + r \left(\dot{\phi} \sin \theta \hat{\mathbf{e}}_r + \dot{\phi} \cos \theta \hat{\mathbf{e}}_\theta - \dot{\theta} \hat{\mathbf{e}}_\phi \right) \times \hat{\mathbf{e}}_r \\ &= \dot{r} \hat{\mathbf{e}}_r + r(0) + r \left(\dot{\phi} \sin \theta (\mathbf{0}) + \dot{\phi} \cos \theta \hat{\mathbf{e}}_\phi - \dot{\theta} (-\hat{\mathbf{e}}_\theta) \right) \\ &= \dot{r} \hat{\mathbf{e}}_r + r \cos \theta \dot{\phi} \hat{\mathbf{e}}_\phi + r \dot{\theta} \hat{\mathbf{e}}_\theta \end{aligned} \quad (25)$$

Then we can simply use the $\mathcal{S}$ -frame representation of $\mathbf{v}$ to find an expression

for the kinetic energy T :

$$\begin{aligned}
T &= \frac{mv^2}{2} = \frac{m}{2} \mathbf{v}^T \mathbf{v} = \frac{m}{2} (v_r^2 + v_\phi^2 + v_\theta^2) \\
&= \frac{m}{2} \left(\dot{r}^2 + (r \cos \theta \dot{\phi})^2 + (r \dot{\theta})^2 \right) \\
&= \frac{m}{2} \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right)
\end{aligned} \tag{26}$$

So, the Lagrangian can be written as:

$$L = \frac{m}{2} \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right) + \frac{m\mu}{r} \left[1 - J_2 \left(\frac{r_o}{r} \right)^2 P_2(\sin \theta) \right] \tag{27}$$

However, this expression still needs to have substituted in the second degree, zeroth order, Legendre polynomial $P_2(\sin \theta)$, which can be found by the following formula:

$$P_n(x) = \sum_{m=0}^M (-1)^m \frac{(2n-2m)!}{2^n m! (n-m)! (n-2m)!} x^{n-2m} \tag{28}$$

Where n is degree and $M = n/2$.

Evaluating it becomes:

$$\begin{aligned}
P_2(\sin \theta) &= \sum_{m=0}^1 (-1)^m \frac{(2(2)-2m)!}{2^2 m! (2-m)! (2-2m)!} \sin^{2-2m} \theta \\
&= (-1)^0 \frac{(4-0)!}{2^2 (0)! (2-0)! (2-0)!} \sin^2 \theta + (-1)^1 \frac{(4-2)!}{2^2 (1)! (2-1)! (2-2)!} \sin^0 \theta \\
&= \frac{24}{4 \cdot 1 \cdot 2 \cdot 2} \sin^2 \theta - \frac{2}{4 \cdot 1 \cdot 1 \cdot 1} \\
&= \frac{3}{2} \sin^2 \theta - \frac{1}{2}
\end{aligned} \tag{29}$$

So then substituting into the found Lagrangian, we get its proper form:

$$\begin{aligned}
L &= \frac{m}{2} \left(\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2 \right) + \frac{m\mu}{r} \left(1 - J_2 \left(\frac{r_o}{r} \right)^2 \left(\frac{3}{2} \sin^2 \theta - \frac{1}{2} \right) \right) \\
&= \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{m\mu J_2 r_o^2}{r^3} \left(\frac{3}{2} \sin^2 \theta - \frac{1}{2} \right) \\
&= \frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3}
\end{aligned} \tag{30}$$

(b.2): Find

Derive the generalized force for Eq. (2) associated with $\mathbf{q} = [r, \phi, \theta]^T$.

Analysis/Conclusions:

In part (a.2), $\mathbf{q} = \mathbf{r}$ made the Jacobian in $\mathbf{Q} = \mathbf{F}_D^T \frac{\partial \mathbf{r}}{\partial \mathbf{q}}$ the identity matrix, but now that $\mathbf{q} = [r, \phi, \theta]^T$, we must find how the position changes with respect to r , ϕ , and θ .

Let us consider the total derivative of $\mathbf{r}$ from the perspective of the chain rule:

$$\frac{d\mathbf{r}}{dt} = \frac{\partial \mathbf{r}}{\partial r} \dot{r} + \frac{\partial \mathbf{r}}{\partial \phi} \dot{\phi} + \frac{\partial \mathbf{r}}{\partial \theta} \dot{\theta} = \frac{\partial \mathbf{r}}{\partial \mathbf{q}} \dot{\mathbf{q}} \quad (31)$$

Then put it side by side with the expression found for $\mathbf{v} = \frac{d\mathbf{r}}{dt}$:

$$\mathbf{v} = \hat{\mathbf{e}}_r \dot{r} + r \cos \theta \hat{\mathbf{e}}_\phi \dot{\phi} + r \hat{\mathbf{e}}_\theta \dot{\theta} \quad (32)$$

Thus from this we see that:

$$\frac{\partial \mathbf{r}}{\partial r} = \hat{\mathbf{e}}_r, \quad \frac{\partial \mathbf{r}}{\partial \phi} = r \cos \theta \hat{\mathbf{e}}_\phi, \quad \frac{\partial \mathbf{r}}{\partial \theta} = r \hat{\mathbf{e}}_\theta \quad (33)$$

So when we compute the generalized force using $Q_j = \mathbf{F}_D \cdot \frac{\partial \mathbf{r}}{\partial q_j}$

$$\begin{aligned} Q_r &= \mathbf{F}_D \cdot \frac{\partial \mathbf{r}}{\partial r} = \left(-\frac{1}{2} C_D A \rho(r) v \mathbf{v} \right) \cdot \hat{\mathbf{e}}_r \\ &= -\frac{1}{2} C_D A \rho(r) v \left(\hat{\mathbf{e}}_r \dot{r} + r \cos \theta \hat{\mathbf{e}}_\phi \dot{\phi} + r \hat{\mathbf{e}}_\theta \dot{\theta} \right) \cdot \hat{\mathbf{e}}_r \\ &= -\frac{1}{2} C_D A \rho(r) v \dot{r} \\ Q_\phi &= \mathbf{F}_D \cdot \frac{\partial \mathbf{r}}{\partial \phi} = \left(-\frac{1}{2} C_D A \rho(r) v \mathbf{v} \right) \cdot (r \cos \theta \hat{\mathbf{e}}_\phi) \\ &= -\frac{1}{2} C_D A \rho(r) v \left(\hat{\mathbf{e}}_r \dot{r} + r \cos \theta \hat{\mathbf{e}}_\phi \dot{\phi} + r \hat{\mathbf{e}}_\theta \dot{\theta} \right) \cdot (r \cos \theta \hat{\mathbf{e}}_\phi) \\ &= -\frac{1}{2} C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi} \\ Q_\theta &= \mathbf{F}_D \cdot \frac{\partial \mathbf{r}}{\partial \theta} = \left(-\frac{1}{2} C_D A \rho(r) v \mathbf{v} \right) \cdot (r \hat{\mathbf{e}}_\theta) \\ &= -\frac{1}{2} C_D A \rho(r) v \left(\hat{\mathbf{e}}_r \dot{r} + r \cos \theta \hat{\mathbf{e}}_\phi \dot{\phi} + r \hat{\mathbf{e}}_\theta \dot{\theta} \right) \cdot (r \hat{\mathbf{e}}_\theta) \\ &= -\frac{1}{2} C_D A \rho(r) v r^2 \dot{\theta} \end{aligned} \quad (34)$$

Which makes the generalized force:

$$\mathbf{Q} = -\frac{1}{2}C_D A \rho(r) v \left(\dot{r} \hat{e}_r + r^2 \cos^2 \theta \dot{\phi} \hat{e}_\phi + r^2 \dot{\theta} \hat{e}_\theta \right) \quad (35)$$

(b.3): Find

Derive the Lagrange's equations and write the result as $\ddot{r}=?$, $\ddot{\phi}=?$, and $\ddot{\theta}=?$.

Analysis/Conclusions:

Once again this system has a minimal coordinate set and a non-conservative force from drag. So, the general form of Lagrange's equations with non-conservative general force may be used:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial L}{\partial \mathbf{q}} = \mathbf{Q}_{\text{nc}} \quad (36)$$

First, we evaluate the generalized momentum terms:

$$\begin{aligned} \frac{\partial L}{\partial \dot{\phi}} &= \frac{\partial}{\partial \dot{\phi}} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\ &= 0 + \frac{mr^2 \cos^2 \theta \partial \dot{\phi}^2}{2 \partial \dot{\phi}} + 0 + 0 - 0 + 0 = mr^2 \dot{\phi} \cos^2 \theta \end{aligned} \quad (37)$$

And the θ momentum:

$$\begin{aligned} \frac{\partial L}{\partial \dot{\theta}} &= \frac{\partial}{\partial \dot{\theta}} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\ &= 0 + 0 + \frac{mr^2 \partial \dot{\theta}^2}{2 \partial \dot{\theta}} + 0 - 0 + 0 = mr^2 \dot{\theta} \end{aligned} \quad (38)$$

And the r momentum:

$$\begin{aligned} \frac{\partial L}{\partial \dot{r}} &= \frac{\partial}{\partial \dot{r}} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\ &= \frac{m}{2} \frac{\partial \dot{r}^2}{\partial \dot{r}} + 0 + 0 + 0 - 0 + 0 = m\dot{r} \end{aligned} \quad (39)$$

Then we compute their time derivatives:

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) &= \frac{d}{dt} (mr^2 \dot{\phi} \cos^2 \theta) \\
&= m \left(\dot{\phi} \cos^2 \theta \frac{dr^2}{dt} + r^2 \cos^2 \theta \frac{d\dot{\phi}}{dt} + r^2 \dot{\phi} \frac{d}{dt} \cos^2 \theta \right) \\
&= m \left(\dot{\phi} \cos^2 \theta \frac{dr^2}{dr} \frac{dr}{dt} + r^2 \cos^2 \theta \frac{d\dot{\phi}}{dt} + r^2 \dot{\phi} \frac{d}{d\theta} (\cos^2 \theta) \frac{d\theta}{dt} \right) \quad (40) \\
&= m \left(2r\dot{r}\dot{\phi} \cos^2 \theta + r^2 \ddot{\phi} \cos^2 \theta - 2r^2 \dot{\phi} \dot{\theta} \cos \theta \sin \theta \right) \\
&= mr \cos \theta \left(2\dot{r}\dot{\phi} \cos \theta + r\ddot{\phi} \cos \theta - 2r\dot{\phi}\dot{\theta} \sin \theta \right)
\end{aligned}$$

And the θ time derivative:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = \frac{d}{dt} (mr^2 \dot{\theta}) = m \left(\frac{dr^2}{dr} \frac{dr}{dt} \dot{\theta} + r^2 \frac{d\dot{\theta}}{dt} \right) = 2mr\dot{r}\dot{\theta} + mr^2 \ddot{\theta} = mr (2\dot{r}\dot{\theta} + r\ddot{\theta}) \quad (41)$$

And the r time derivative:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) = m \frac{dr}{dt} = m\ddot{r} \quad (42)$$

Since the Lagrangian is not a function of ϕ :

$$\frac{\partial L}{\partial \phi} = 0 \quad (43)$$

Then compute the θ derivative:

$$\begin{aligned}
\frac{\partial L}{\partial \theta} &= \frac{\partial}{\partial \theta} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\
&= 0 + \frac{mr^2 \dot{\phi}^2}{2} \frac{\partial}{\partial \theta} (\cos^2 \theta) + 0 + 0 - \frac{3m\mu J_2 r_o^2}{2r^3} \frac{\partial}{\partial \theta} (\sin^2 \theta) + 0 \\
&= -mr^2 \dot{\phi}^2 \cos \theta \sin \theta - 3m\mu J_2 r_o^2 r^{-3} \cos \theta \sin \theta \\
&= -m \cos \theta \sin \theta \left(r^2 \dot{\phi}^2 + \frac{3\mu J_2 r_o^2}{r^3} \right) \quad (44)
\end{aligned}$$

And the r derivative:

$$\begin{aligned}
&= \frac{\partial L}{\partial r} \\
&= \frac{\partial}{\partial r} \left(\frac{m\dot{r}^2}{2} + \frac{mr^2 \cos^2 \theta \dot{\phi}^2}{2} + \frac{mr^2 \dot{\theta}^2}{2} + \frac{m\mu}{r} - \frac{3m\mu J_2 r_o^2}{2r^3} \sin^2 \theta + \frac{m\mu J_2 r_o^2}{2r^3} \right) \\
&= 0 + \frac{m \cos^2 \theta \dot{\phi}^2}{2} \frac{dr^2}{dr} + \frac{m\dot{\theta}^2}{2} \frac{dr^2}{dr} + m\mu \frac{dr^{-1}}{dr} - \frac{3m\mu J_2 r_o^2}{2} \sin^2 \theta \frac{dr^{-3}}{dr} + \frac{m\mu J_2 r_o^2}{2} \frac{dr^{-3}}{dr} \\
&= mr\dot{\phi}^2 \cos^2 \theta + mr\dot{\theta}^2 - m\mu r^{-2} + \frac{9}{2}m\mu J_2 r_o^2 r^{-4} \sin^2 \theta - \frac{3}{2}m\mu J_2 r_o^2 r^{-4} \\
&= mr \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \mu r^{-3} + \frac{9}{2}\mu J_2 r_o^2 r^{-5} \sin^2 \theta - \frac{3}{2}\mu J_2 r_o^2 r^{-5} \right) \\
&= mr \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3\mu J_2 r_o^2}{2r^5} (3 \sin^2 \theta - 1) \right)
\end{aligned} \tag{45}$$

So, as per Eq. (36), we can then equate the above calculated terms to their respective non-conservative generalized drag force to get the equations of motion.

For the coordinate ϕ we set up the EoM as:

$$\begin{aligned}
&\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \frac{\partial L}{\partial \phi} = Q_{\phi,nc} \\
&mr \cos \theta \left(2\dot{r}\dot{\phi} \cos \theta + r\ddot{\phi} \cos \theta - 2r\dot{\phi}\dot{\theta} \sin \theta \right) - 0 = -\frac{1}{2}C_D A \rho(r) v r^2 \cos^2 \theta \dot{\phi}
\end{aligned} \tag{46}$$

And we then isolate the $\ddot{\phi}$ term:

$$\begin{aligned}
\ddot{\phi} m r^2 \cos^2 \theta &= -\frac{1}{2}C_D A \rho(r) v \dot{\phi} r^2 \cos^2 \theta - 2\dot{r}\dot{\phi} m r \cos^2 \theta + 2\dot{\phi}\dot{\theta} \sin \theta m r^2 \cos \theta \\
\ddot{\phi} &= -\frac{1}{2}C_D A \rho(r) v \dot{\phi} \frac{r^2 \cos^2 \theta}{m r^2 \cos^2 \theta} - 2\dot{r}\dot{\phi} \frac{m r \cos^2 \theta}{m r^2 \cos^2 \theta} + 2\dot{\phi}\dot{\theta} \frac{\sin \theta}{\cos \theta} \frac{m r^2 \cos \theta}{m r^2 \cos \theta} \\
\ddot{\phi} &= -\frac{C_D A \rho(r) v \dot{\phi}}{2m} - \frac{2\dot{r}\dot{\phi}}{r} + 2\dot{\phi}\dot{\theta} \tan \theta
\end{aligned} \tag{47}$$

For the coordinate θ we set up the EoM as:

$$\begin{aligned}
&\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = Q_{\theta,nc} \\
&mr \left(2\dot{r}\dot{\theta} + r\ddot{\theta} \right) + m \cos \theta \sin \theta \left(r^2 \dot{\phi}^2 + \frac{3\mu J_2 r_o^2}{r^3} \right) = -\frac{1}{2}C_D A \rho(r) v r^2 \dot{\theta}
\end{aligned} \tag{48}$$

And we then isolate the $\ddot{\theta}$ term:

$$\begin{aligned}
\ddot{\theta} m r^2 &= -\frac{1}{2} C_D A \rho(r) v \dot{\theta} r^2 - 2 \dot{r} \dot{\theta} m r - \dot{\phi}^2 \cos \theta \sin \theta m r^2 - 3 \mu J_2 r_o^2 \cos \theta \sin \theta \frac{m}{r^3} \\
\ddot{\theta} &= -\frac{1}{2} C_D A \rho(r) v \dot{\theta} \frac{r^2}{m r^2} - 2 \dot{r} \dot{\theta} \frac{m r}{m r^2} - \dot{\phi}^2 \cos \theta \sin \theta \frac{m r^2}{m r^2} - 3 \mu J_2 r_o^2 \cos \theta \sin \theta \frac{m}{r^3} \frac{1}{m r^2} \\
\ddot{\theta} &= -\frac{C_D A \rho(r) v \dot{\theta}}{2m} - \frac{2 \dot{r} \dot{\theta}}{r} - \dot{\phi}^2 \cos \theta \sin \theta - \frac{3 \mu J_2 r_o^2}{r^5} \cos \theta \sin \theta \\
\ddot{\theta} &= -\frac{C_D A \rho(r) v \dot{\theta}}{2m} - \frac{2 \dot{r} \dot{\theta}}{r} - \cos \theta \sin \theta \left(\dot{\phi}^2 + \frac{3 \mu J_2 r_o^2}{r^5} \right)
\end{aligned} \tag{49}$$

For the coordinate r we set up the EoM as:

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) - \frac{\partial L}{\partial r} &= Q_{r,nc} \\
m \ddot{r} - m r \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3 \mu J_2 r_o^2}{2 r^5} (3 \sin^2 \theta - 1) \right) &= -\frac{1}{2} C_D A \rho(r) v \dot{r}
\end{aligned} \tag{50}$$

And we then isolate the $\ddot{r}$ term:

$$\begin{aligned}
\ddot{r} m &= -\frac{1}{2} C_D A \rho(r) v \dot{r} + r \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3 \mu J_2 r_o^2}{2 r^5} (3 \sin^2 \theta - 1) \right) m \\
\ddot{r} &= -\frac{1}{2} C_D A \rho(r) v \dot{r} \frac{1}{m} + r \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3 \mu J_2 r_o^2}{2 r^5} (3 \sin^2 \theta - 1) \right) \frac{m}{m} \\
\ddot{r} &= -\frac{C_D A \rho(r) v \dot{r}}{2m} + r \left(\dot{\phi}^2 \cos^2 \theta + \dot{\theta}^2 - \frac{\mu}{r^3} + \frac{3 \mu J_2 r_o^2}{2 r^5} (3 \sin^2 \theta - 1) \right) \\
\ddot{r} &= -\frac{C_D A \rho(r) v \dot{r}}{2m} + r \dot{\phi}^2 \cos^2 \theta + r \dot{\theta}^2 - \frac{\mu}{r^2} + \frac{3 \mu J_2 r_o^2}{2 r^4} (3 \sin^2 \theta - 1)
\end{aligned} \tag{51}$$

Which makes the EoM:

$$\begin{aligned}
\ddot{r} &= -\frac{C_D A \rho(r) v \dot{r}}{2m} + r \dot{\phi}^2 \cos^2 \theta + r \dot{\theta}^2 - \frac{\mu}{r^2} + \frac{3 \mu J_2 r_o^2}{2 r^4} (3 \sin^2 \theta - 1) \\
\ddot{\theta} &= -\frac{C_D A \rho(r) v \dot{\theta}}{2m} - \frac{2 \dot{r} \dot{\theta}}{r} - \cos \theta \sin \theta \left(\dot{\phi}^2 + \frac{3 \mu J_2 r_o^2}{r^5} \right) \\
\ddot{\phi} &= -\frac{C_D A \rho(r) v \dot{\phi}}{2m} - \frac{2 \dot{r} \dot{\phi}}{r} + 2 \dot{\phi} \dot{\theta} \tan \theta
\end{aligned} \tag{52}$$

Where: $v = \sqrt{\dot{r}^2 + r^2 \cos^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2}$

(b.4): Find

For zero drag, identify the cyclic coordinate and the associated conserved generalized momentum. Discuss its relation to the angular momentum about $\hat{e}_z$, denoted by H_z .

Analysis/Conclusions:

It was identified in (b.3) that the Lagrangian is not a function of ϕ . **This means that ϕ is the cyclic coordinate.**

If the force of drag is removed, this removes the only non-conservative force, and the Lagrange equation for ϕ now evaluates as:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \frac{\partial L}{\partial \phi} = 0 \Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = 0 \quad (53)$$

Because $\frac{\partial L}{\partial \phi} = 0$ as determined in (b.3), the generalized momentum p_ϕ is constant over time.

Likewise, the equation for this constant momentum was also found in (b.3):

$$p_\phi = \frac{\partial L}{\partial \dot{\phi}} = mr^2 \dot{\phi} \cos^2 \theta \quad (54)$$

Which can be evaluated from θ and $\dot{\phi}$ at any time.

THEN SOMETHING SOMETHING H_z IDK